

## 3.2 Understanding Portfolio Risk



### Guiding Questions

1. What is the difference between total risk, systematic risk, and unsystematic risk?
2. Why can diversification reduce some types of risk but not others?
3. How does beta measure a portfolio's sensitivity to market movements?



# EXPLORE

### What will you learn?

- How risk reflects uncertainty in investment outcomes.
- The difference between systematic risk and unsystematic risk.
- Why diversification reduces firm-specific risk.
- How beta measures exposure to overall market risk.
- Why understanding risk is essential for evaluating investment performance.

#### 1. Portfolio risk

The most common measure of portfolio risk is standard deviation ( $\sigma$ ).

Standard deviation measures how much returns fluctuate around their average value.

- High standard deviation → returns vary widely → high risk
- Low standard deviation → returns are more stable → lower risk

Standard deviation captures total portfolio risk, including all sources of volatility.

## Comparing Portfolio Risk

Consider two portfolios with the same average return.

### Portfolio Average Return Standard Deviation

|   |     |     |
|---|-----|-----|
| A | 10% | 5%  |
| B | 10% | 15% |

### Interpretation

Both portfolios produce the same return, but Portfolio B is much more volatile.

Therefore:

- Portfolio B is riskier because its returns fluctuate more widely.

Risk-averse investors typically prefer lower volatility for the same return.

## 2. Types of Portfolio Risk

Portfolio risk can be divided into **two major categories**.

### Unsystematic Risk (Diversifiable Risk)

Unsystematic risk arises from firm-specific or industry-specific factors, such as:

- management decisions
- product failures
- labour strikes
- company scandals

These events affect individual firms rather than the whole market.

Because these risks are independent across companies, they can be reduced through diversification.

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### Example

Suppose an investor owns shares only in an airline company.

A sudden increase in oil prices may severely affect that company.

If the investor also holds stocks in:

- technology companies
- banks
- healthcare firms

the overall portfolio impact will be much smaller.

This illustrates how diversification reduces unsystematic risk.

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### 3. Systematic Risk (Market Risk)

Systematic risk refers to economy-wide factors that affect all assets.

Examples include:

- economic recessions
- inflation shocks
- interest rate changes
- global financial crises

Because these factors affect all firms simultaneously, they cannot be eliminated through diversification.

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### 4. Measuring Systematic Risk with Beta

Systematic risk is measured using beta ( $\beta$ ).

Beta measures how sensitive a portfolio's returns are to movements in the overall market.

Interpretation:

#### Beta Meaning

$\beta = 1$  Portfolio moves with the market

$\beta > 1$  Portfolio is more volatile than the market

$\beta < 1$  Portfolio is less volatile than the market

#### Interpreting Beta

Suppose the market increases by 10%.

#### Portfolio Beta Expected Change

A            1.0    about 10%

B            1.5    about 15%

## Portfolio Beta Expected Change

C      0.7    about 7%

Portfolio B is more sensitive to market movements, meaning it carries higher systematic risk.

## 5. Diversification and Portfolio Risk

Diversification works by combining assets whose risks are not perfectly correlated.

As the number of assets increases:

- Unsystematic risk declines
- Systematic risk remains

Graphically, portfolio risk falls quickly as diversification increases, but eventually levels off at market risk.

This means investors cannot eliminate all risk, only firm-specific risk.

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## Key Takeaways

From this section we learn:

- Portfolio risk consists of systematic and unsystematic components.
- Diversification reduces unsystematic risk.
- Beta measures exposure to market risk.

Understanding these concepts is essential before evaluating risk-adjusted performance, which is introduced in the next section.

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## Reflect

1. What does risk mean in the context of investment portfolios?
2. What is the difference between systematic and unsystematic risk?
3. Why can diversification eliminate unsystematic risk but not systematic risk?
4. What does a beta of 1.4 indicate about a portfolio?

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## Solve

### Question 1

Two portfolios have the following characteristics:

**Portfolio Average Return Standard Deviation**

|   |    |     |
|---|----|-----|
| A | 9% | 8%  |
| B | 9% | 18% |

Which portfolio would a risk-averse investor prefer? Explain.

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**Question 2**

A portfolio has a beta of 1.3.

If the market increases by 10%, what approximate change would you expect in the portfolio's return?

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**Question 3**

Explain how holding stocks in multiple industries helps reduce portfolio risk.

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Answers available [here \(https://learn.usc.edu.au/courses/35359/files/3292595?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3292595?wrap=1).

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**Critically evaluate**

Suppose an investor holds 100 different stocks, yet their portfolio still declines significantly during a global recession.

Explain why diversification did not eliminate this loss.